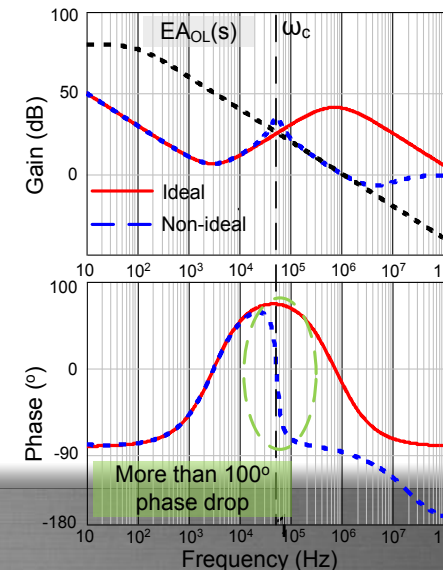


Homework 4

- 1. Follow the Voltage-Mode Controlled Buck Converter you designed in HW3.
- (1) (15%) Define the spec. of amplifier including dc gain and 3dB bandwidth first, which achieves designed compensation transfer function and loop phase margin in HW3. Then use simulation tool to verify. Hint: Use level 2 model of amplifier to simulate compensation transfer function with external compensation components and loop gain. Please also show the test circuit for verification.



Homework 4 (2)

- (2) (35%) Please design the transistor level of the amplifier used for compensator. **Use the provided FinFET process.** Please write down the design procedure. Then use simulation tool to verify that your design meets the spec. of amplifier in (1).
- You get extra points if you compare the simulated load transient responses, loop gains, and compensator gains of designed Voltage-Mode Controlled Buck Converter using level 1, 2, 3 models of amplifier.

(6% for each comparison item. Multiply by 1 (2) if you compare level 1, 2, (3))

level 1: behavior model (ideal). level 2: behavior model (finite dc gain and BW). level 3: transistor level model

